VERA: Generating Visual Explanations of Two-Dimensional Embeddings via Region Annotation

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***Abstract*—Two-dimensional embeddings obtained from dimensionality reduction techniques ~~,~~ such as MDS, t-SNE, ~~and~~ or UMAP, are widely used ~~across various disciplines~~ to visualize high-dimensional data ~~. These visualizations are especially valuable during exploratory data analysis, allowing researchers to visually identify~~ and support researchers in visually identifying clusters, outliers, and other interesting patterns in the data. However, interpreting and contextualizing the identified patterns ~~can be challengingand often requires additional~~ remains challenging, often requiring manual inspection which can be both time-consuming and repetitive. ~~To address this issue, we propose~~ We propose VERA (Visual Explanations via Region Annotation~~(VERA), an automated, static embedding-annotation approach that generates~~), a general-purpose method that generates static visual explanations for two-dimensional embeddings. VERA identifies informative regions in the embedding space and ~~automatically produces visual explanations using provided~~ associates them with human-interpretable features, ~~allowing users to obtain a high-level overview~~ producing concise visual annotations that summarize the structure of the embedding landscape at a glance. We ~~illustrate the usage of VERA~~demonstrate VERA's utility on several real-world datasets and ~~validate the effectiveness of our approach with~~ evaluate its effectiveness in a user study ~~, comparing the static explanations produced by VERA to~~ comparing it with the utility of a comprehensive, interactive data-mining toolkit. ~~Our results demonstrate that the static explanations generated by VERA are as useful as those obtained from interactive tools on~~ Study participants who used VERA achieved comparable accuracy on typical exploratory analysis tasks while requiring significantly less time and effort. Our results show that VERA's generated static explanations can convey the essential insights of complex embeddings and support users in typical exploratory data analysis tasks~~but require~~ , while requiring significantly less time and effort from the user.

***Index Terms*—Dimensionality reduction, Embedding, Visualization, Visual explanation, Explainable AI**

I. Introduction

TWO-DIMENSIONAL, point-based visualizations of high-dimensional data are ~~ubiquitous throughout various domains of science, including~~ a cornerstone of exploratory data analysis and are used across diverse scientific domains such as biology, economics, chemistry, political science, astronomy, and physics [1]. ~~With the advent of big data , analyzing data~~ As data sets with tens of thousands of features and millions of samples ~~has now become a commonplaceoccurrence. A common step during exploratory data analysis is the visualization of these~~ have become commonplace, visual exploration is often the first step toward discovering meaningful patterns and generating hypotheses. Such visualizations, often called *embeddings*, project high-dimensional ~~data sets.~~ relationships into a two-dimensional space, as illustrated in Fig. 1. Once the embedding is obtained, analysts seek to interpret what its shapes and clusters represent in the original data.

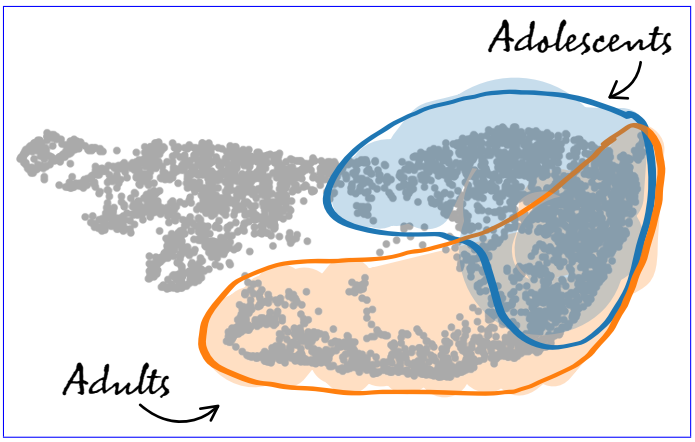


Fig. 1. An example of a two-dimensional embedding and its possible interpretation. The UMAP visualization shows the relationships between 4,177 marine mollusks. After obtaining the embedding, the next step is to determine what the shapes and clusters represent. For example, we might determine that the mollusks in the blue region predominantly correspond to adolescent animals while those in the orange region correspond to mature mollusks.

Today, data science practitioners have at their disposal a plethora of scalable dimensionality reduction methods that can be used to reduce data to two dimensions, including principal components analysis (PCA) [2], linear discriminant analysis (LDA) [3], FreeViz [4], multidimensional scaling (MDS) [5], t-distributed stochastic neighbor embedding (t-SNE) [6], and uniform manifold approximation and projection (UMAP) [7]. These ~~are routinely used to visually uncover underlying structures, such as~~ methods are routinely applied to reveal clusters, transitions, ~~or outliers, and to support exploratory dataanalysis and hypothesis generation~~and outliers that provide insight into the underlying data's landscape.

After performing dimensionality reduction, analysts typically examine the resulting visualization to identify any discernible visual patterns and interpret how these reflect un-

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derlying phenomena in the data. Linear methods, such as PCA, LDA, and FreeViz, offer intuitive interpretations since their inferred dimensions correspond to linear combinations of the original features. However, these are often too limited in the types of high-dimensional structures they can reveal. Nonlinear methods, such as MDS, t-SNE, and UMAP, can reveal more complex patterns and relationships, but relating the obtained embedding dimensions to the original features becomes more challenging. Moreover, with the development of increasingly yet the lack of direct correspondence between embedding dimensions and original features makes interpretation difficult. As scalable dimensionality reduction techniquesalgorithms continue to improve [7]–[9], analysts often frequently generate multiple embeddings using different parameter settings and or algorithms, each potentially revealing different revealing distinct aspects of the underlying data manifold [10]. Each resulting embedding must then be inspected and interpreted, a task process that is both time-consumingand repetitive, subjective, and cognitively demanding.

Existing research on interpreting two-dimensional embeddings has mostly focused on developing comprehensive interactive tools that help analysts explore every facet of resulting embeddings focused primarily on interactive visualization tools that allow users to explore the embedding space in detail. Examples include systems that link embeddings to feature distributions, enable brushing and linking, or provide cluster exploration interfaces (e.g., [11], [12]). While these tools enable a detailed exploration of each particular embedding, the primary goal of EDA an in-depth analysis, the main goal of exploratory data analysis (EDA) is often to obtain a concise, high-level overview of the recovered embedding structure rather than a detailed investigation of each visual pattern . to investigate every local pattern interactively.

Although many tools offer features to speed up and assist in the exploration process, e.g., automatic clustering or feature ranking, analysts must still actively navigate and make sense of the embedding, which is both cumbersome and time-consuming. activities that are both repetitive and time-intensive. A typical workflow might may involve selecting a group of data points, inspecting their feature distributions, comparing these to the remaining data points, and annotating the visualization accordingly. This must, of course, be repeated for each interesting group of data points in each resulting embedding.

Given the repetitive nature of this task, well-designed these exploratory data analysis tasks, automated, static approaches that require no user interaction can both speed up the interpretation process and reduce the cognitive load on analysts. in explaining the data landscapes can substantially reduce analysis time and cognitive effort while still conveying the essential structure of the embedding. Such methods can complement interactive tools by providing quick, interpretable summaries that guide deeper investigation when needed.

In this work, we propose *Visual Explanations via Region Annotation (VERA)*, an automated, general-purpose embedding annotation tool method for generating static visual explanations of two-dimensional embeddings. Similarly to other general-purpose approaches, VERA operates on datasets with at least some number of VERA automatically identifies informative regions in the embedding space and associates them with human-interpretable features, among which VERA automatically identifies and highlights features associated with distinct regions of the embedding , presenting them in a features, presenting these associations as concise, small-multiple fashionvisual summaries. This approach helps analysts quickly and easily obtain a high-level overview of the obtained embedding. While a handful of static explanation approaches

While several static explanation methods have already been proposed, these are often limited to a single explanation [13]–[15]or require the user to sift through a large number of output visualizations [16], [17]. A single explanation is often insufficient for explaining all relevant visual structures, while requiring users to inspect [13]–[15], existing approaches either produce a single global explanation—often insufficient to capture multiple meaningful structures or generate a large number of visualizations diminishes much of their efficiency over interactive tools. In contrast, VERA generates that users must manually inspect [16], [17]. VERA overcomes these limitations by generating a small set of highly informative visualizations, retaining the speed and simplicity of static visualizations without sacrificing the informativeness of interactive approaches in common exploratory data analysis tasks. To validate our approach, we conducted a

Our main contributions are as follows:

1) We introduce VERA, a general-purpose framework for generating region-based, static visual explanations of two-dimensional data embeddings.
2) We present an automated procedure for detecting informative regions of point-based visualizations and linking them to relevant, human-interpretable features.
3) We validate VERA through a user study comparing VERA to interactive tools for embedding interpretation and found that users given explanations generated by VERA performed as well as users with access to comprehensive, interactive tools on several common, high-level it with the utility of a comprehensive interactive data-mining toolkit, showing that VERA enables users to perform typical embedding interpretation tasks with comparable accuracy while requiring significantly less time.

Together, these contributions demonstrate that VERA's static visual explanations can effectively convey the essential insights of complex embeddings, offering a fast and accessible complement to interactive data exploration.

II. Related Work

Alongside the development of novel dimensionality reduction approaches for the visualization of high-dimensional data [5]–[7], numerous tools have been developed to aid in assessing the qualityof the obtained embeddingsevaluating

embedding quality [18]–[20], interpreting ~~the resulting~~ visual structures [21]–[23], and comparing multiple embeddings [24]–[26].

Most embedding interpretation approaches aim to relate the obtained embedding to the original, high-dimensional features. General-purpose techniques implicitly assume that the original features are, to some degree, semantically meaningful and human-interpretable. In domains where this is not the case – such as raw image pixels, compositional data, or deep latent representations – domain-specific solutions are often required, as in the case ~~for~~ of text [27], [28], images [29], [30], and drug-prescription data [31].

General-purpose interpretation techniques can be broadly divided into global and local approaches. Global approaches typically aim to express the embedding axes as linear combinations of the original features [32], [33]. While effective for linear projections, most modern dimensionality reduction techniques rely on highly ~~non-linear~~ nonlinear transformations of the original feature space, making global approaches unsuitable for interpreting these embeddings. Local approaches instead aim to explain local visual patterns such as clusters, transitions, and outliers and relate them to particular combinations of high-dimensional feature values. These methods typically enrich the resulting two-dimensional scatter plot with additional graphical elements or overlays [15], [21], [34] or provide an interactive interface through which users can navigate and explore the embedding space [19], [35], [36]. Broadly speaking, local approaches can be categorized into two types: descriptive approaches and contrastive approaches.

Descriptive approaches identify regions of the embedding space, most commonly clusters, and determine which combinations of high-dimensional feature values are characteristic of the samples contained within each region. Researchers have proposed several strategies to identify such feature combinations. For instance, Joia *et al.* [34] identify salient features by examining the SVD loadings associated with each visual cluster, while Da Silva *et al.* [13] and subsequent extensions by Van Driel *et al.* [14] and Tian *et al.* [15] adopt a variance-based approach, identifying features that exhibit locally low variance within different embedding regions. A similar interactive, variance-based approach was proposed by Pagliosa *et al.* [37]. Alternatively, some methods embed both samples and features into a shared embedding space, enabling the identification of characteristic features based on their proximity to samples in the embedding space. For instance, Broeksema *et al.* [32] use Multiple Correspondence Analysis (MCA) to jointly embed samples and features, then identify characteristic regions using Voronoi tessellations. Cheng and Mueller [21] developed a conceptually similar approach using MDS for joint embedding and kernel density estimation for region definition.

While descriptive approaches help explain *what* particular regions correspond to, contrastive approaches aim to explain ~~how a given region *differs* from other regions~~*how* one region differs from others. In a recent survey of sociological literature, Miller [38] highlights that humans prefer contrastive explanations and concludes that explanations in artificial intelligence should account for this preference. Contrastive approaches generally aim to identify combinations of feature values that distinguish one region from another, and a variety of techniques have been proposed to this end. Several approaches combine well-established statistical tools – such as feature distribution plots [19], [39], statistical tests [22], decision trees [36], and Shapley values [40] – with intuitive, interactive user interfaces. Others propose custom solutions; for example, Fujiwara *et al.* [23] propose a contrastive variant of PCA, while Kandogan [35] combines several hand-crafted metrics.

Despite the plethora of existing approaches, current solutions share several notable limitations. ~~The vast majority of these are designed~~ Most are implemented as interactive tools ~~, requiring a high degree of active~~ that require extensive user interaction, which can be both repetitive and time-consuming, particularly during exploratory data analysis when the goal is to obtain a high-level overview of the embedding. This problem is especially exacerbated when inspecting and comparing multiple embeddings. Other approaches are limited to producing a single explanation, often focusing on the ~~single~~ most salient combination of feature values while neglecting other potentially interesting but less ~~potent~~ prominent combinations. Other approaches are tightly coupled to specific dimensionality reduction techniques, reducing their general applicability across methods [21], [32], [41]. Finally, the majority of existing approaches define regions based on a single, discrete clustering of the embedding space. This approach risks potentially missing overlapping or continuous structures in the embedding space, such as gradual transitions or trajectories, which do not fit neatly within discrete cluster boundaries.

~~To address the challenges highlighted above, we propose VERA , an automatic embedding annotation approach for generating~~ In contrast to existing interpretation methods, VERA automatically produces concise, static visual explanations ~~of arbitrary two-dimensional embeddings. VERA generates multiple contrastive and descriptive explanations of arbitrary two-dimensional embeddings, allowing analysts to get a quick, high-level overview of the obtained embeddingwithout the need to actively explore embeddings in full-fledged graphical interfaces~~without requiring user interaction. It combines descriptive and contrastive perspectives to summarize the embedding's structure, providing informative regional annotations rather than a single global summary or a large set of candidate views. Unlike most existing methods, VERA does not rely on ~~identifying discrete clusters in the embedding space and can identify~~ predefined clusters and can detect and annotate arbitrary visual structures in the embedding space.

III. Methods

Miller [38] posits that the process of explanation involves three cognitive ~~processes~~stages: generation, selection, and evaluation. In the context of explaining two-dimensional embeddings, VERA aims to automate the first ~~and second of thesethree~~two of these. First, ~~VERA~~ it generates several candidate explanations~~– each of which describes a different pattern in~~ , each describing a distinct pattern in the embedding. Second, ~~since some explanations are likely more informative than others, VERA then~~ it ranks and selects the most ~~salient~~

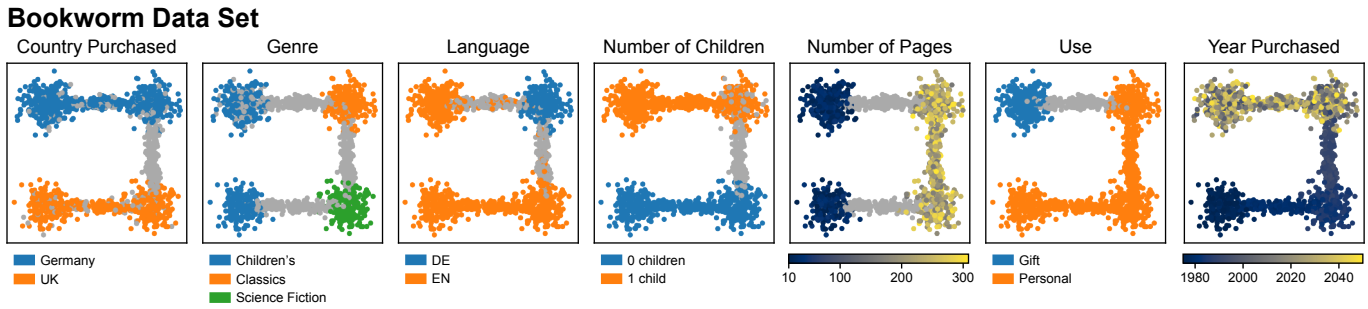


Fig. 2. Scatter plots for each feature of our fictional Bookwork dataset. The point positions correspond to its two-dimensional embedding, revealing four distinct clusters connected by three transitions. Categorical variables are colored with a discrete colormap, while the two numeric variables are colored with a continuous colormap. Missing values are colored gray.

~~explanations~~informative ones. Finally, the selected explanations are shown to the user, ~~with whom the responsibility of~~ who remains responsible for evaluating, interpreting, and contextualizing ~~the explanations ultimately rests~~them.

In line with existing work, we propose two different types of visual explanations: *contrastive explanations* and *descriptive explanations*. Contrastive explanations show how the values of a single, high-dimensional feature ~~are distributed~~ vary across the embedding space. Descriptive explanations, on the other hand, ~~list~~ identify the feature values that ~~describe~~ characterize specific regions of the embedding space, ~~e.g.,~~ such as clusters. Contrastive explanations can help answer questions like "Where do the different feature values occur?" while descriptive explanations answer questions like "What feature values are characteristic for this region?".

Given ~~an arbitrary~~ a data set $D \in \mathbb{R}^{N \times |F|}$ and its corresponding two-dimensional embedding $Y \in \mathbb{R}^{N \times 2}$, our aim is to relate different regions of Y back to the original, high-dimensional features $f \in F$. Some feature values may be characteristic of specific regions of Y; for instance, data points in a particular cluster may all share the same category or contain similar numeric values. We ~~associate groups of informative values into~~ group informative feature values within *rules* ~~, indicating~~ that indicate a particular category or numeric interval. A rule φ_i is defined as a function $\varphi : Y \to \{0, 1\}$, which evaluates to 1 if a sample y satisfies the rule condition and 0 otherwise. Rules are associated with *regions* via *region annotations* e_i, which represent pairs of rules φ_i and regions r_i, $e_i = (r_i, \varphi_i)$. Each region R_i represents a set of points $R_i = \{y_1, y_2, \cdots, y_N\} \subseteq Y$ ~~contained within the~~ belonging to that region. Region annotations $\{e_1, e_2, \cdots, e_k\}$ can be merged into *region annotation groups* e_{group} associated with rule $\varphi_{\text{group}}(y) = \prod_{i=1}^{k} \varphi_i(y)$ and region $R_{\text{group}} = \bigcup_{i=1}^{k} R_i$. Since region annotation groups serve the same conceptual purpose as region annotations, we use the two terms interchangeably. Once we have identified a set of informative region annotations $E = \{e_1, e_2, \cdots, e_K\}$, the next step is to display these to the user. ~~Simultaneously displaying~~ Displaying all region annotations ~~to the user~~ simultaneously in a single visualization would ~~result in~~ lead to overlapping regions and labels, ~~rendering the visualization unreadable~~reducing readability. Instead, we group region annotations into *panels* $P \subseteq E$ (also referred to as *visual explanations*) and arrange collections of panels into *layouts* $L = \{P_1, P_2, \cdots, P_L\}$. A layout is then displayed in a small multiple fashion.

In the upcoming sections, we use a fictional, guiding example to outline the individual steps of our approach. This fictional dataset, aptly named the Bookworm dataset, comprises ~~of~~ samples corresponding to books owned by an imaginary literary enthusiast. Each book is characterized by seven features: ~~book~~ language, genre, number of pages, year and country of purchase, whether ~~or not the book owners~~ the owner had children at the time of purchase, and whether ~~or not~~ the book was purchased for personal use or as a gift. Fig. 2 shows the distribution of feature values as they appear in a specially designed two-dimensional embedding.

Through these books, we can trace the life of this fictional book collector. The bottom left cluster – filled with children's books presumably given to them by their parents – characterizes their childhood. Their teenage years are represented by the bottom right cluster, filled with longer science fiction books. After spending their formative years in the United Kingdom, the collector relocates to Germany, where their literary tastes shift to German classics, which remain their favorites to this day. Soon thereafter, they have their first child, ~~in~~ to whom they resolve to instill an appreciation for the literary arts. However, to preserve their child's English heritage, they limit their offspring to a strict diet of English literature. While this story can be inferred by carefully examining each feature shown in Fig. 2, the following sections will show how VERA produces explanations that uncover this same narrative automatically.

A. Region Construction

Region annotations are the fundamental building blocks of visual explanations, representing groups of data points in the embedding that ~~share a common characteristic~~exhibit shared characteristics. In this section, we outline the steps to identify and construct informative region annotations. We then discuss how to combine these regions into informative visual explanations in the next two sections. At a high level, our region construction procedure ~~follows three steps, depicted~~ consists of three main steps, illustrated in Fig. 3.

1) Extracting Explanatory Variables: An input ~~data set~~ dataset D can include both categorical and numeric features. For categorical features, we apply one-hot encoding,

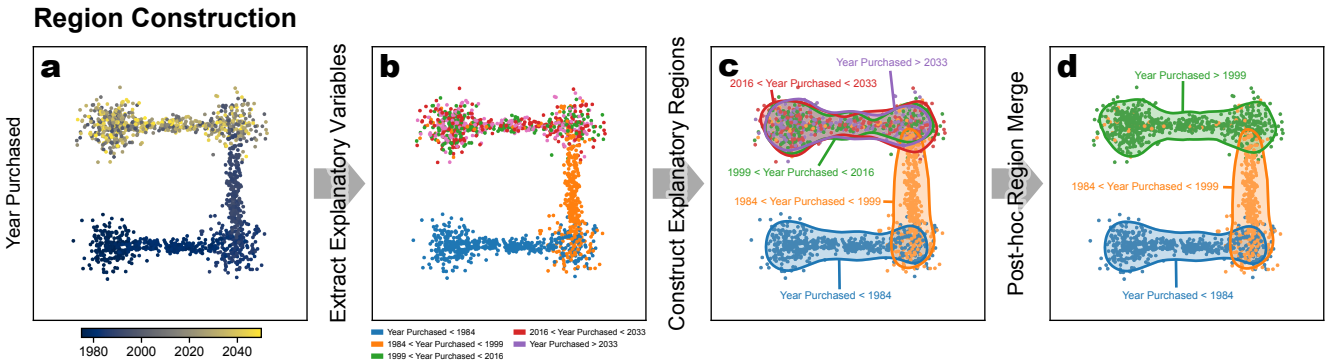


Fig. 3. We illustrate the region construction process for a single feature using a synthetic example. (a) Two-dimensional embeddings are often depicted by coloring points according to their feature values. (b) We discretize the numeric values into five bins, resulting in five binary indicator variables. Here, points are colored according to their discretization bin membership. (c) For each of the discretized bins, we extract regions by tracing the contours of the corresponding KDE. (d) Lastly, we apply our post-hoc merging procedure to consolidate overlapping and semantically compatible region annotations.

~~indicating each sample's membership to a particular category in~~ representing each sample's membership in a given category using k binary indicator variables. For numeric features, we discretize their values into k bins, each represented by a binary indicator variable ~~indicating~~ that denotes whether a sample falls within ~~that bin~~the corresponding interval. For example, the numeric feature in Fig. 3.a is discretized into five bins, with the colors in Fig. 3.b indicating bin membership. We apply these transformations to ~~each~~ every input feature, associating each resulting indicator variable with a rule φ_i that specifies ~~its~~ the corresponding discretization interval or category ~~assignment. The procedure results in~~ membership. This procedure yields a collection of binary indicator variables ~~, which form the basis~~ that serve as the foundation for constructing region annotations.

2) Constructing Region Annotations: Having obtained a collection of binary indicator variables and their associated rules, we next determine ~~where~~ the locations of their corresponding samples ~~are located in~~ within the embedding. For each variable, we obtain its kernel density estimate (KDE), denoted by D, using a Gaussian kernel function. If sample-wise distortion measures of the embedding are provided, we incorporate them as weights during the estimation step. This avoids generating explanations ~~of~~ for poorly mapped samples in the embedding space[1]. Next, we extract contour lines of the KDE at a user-specified level and use the resulting polygons as ~~our~~ the visual representation of a region r_i. By default, we extract contour lines at ~~$0.25 \cdot \max D$~~ $0.25 \times \max D$. By construction, each resulting region encompasses samples that predominantly share a particular feature value. In our running example in Fig. 3, the five categories corresponding to discretization bins in Fig. 3.b are used to construct the five corresponding regions shown in Fig. 3.c. We combine the obtained rules and regions into region annotations $e_i = (r_i, \varphi_i)$, which serve as the building blocks for constructing visual explanations.

An important consideration of the KDE step is ~~setting the~~ the choice of Gaussian kernel bandwidth σ. Different embedding algorithms produce embeddings that can span across different scales, e.g., one embedding may span from -1 to 1 along its dimensions, while another may span from -10 to 10. Additionally, a single scale parameter is unlikely to be appropriate for ~~data sets~~ datasets containing different numbers of samples. For instance, in a dataset with 50 samples, we are likely to be interested in groups of 10-20 points, while in a dataset containing 2 million samples, the groups of interest are likely larger. By default, we set σ to equal the median distance of each point's k-th nearest neighbor, setting $k = \sqrt{N}$. Intuitively, the scale parameter controls the granularity of the ~~obtained~~ resulting region annotations: ~~small values lead to a larger number of smaller regions revealing~~ smaller values produce many fine-grained ~~details, while large values generate larger~~ regions, ~~resulting in more high-level explanations~~whereas larger values yield broader, higher-level regions.

3) Post-hoc Region Merging: When constructing explanatory variables, we assign each sample to one of the resulting binary indicator variables, ~~indicating its membership to~~ denoting its membership in a particular category or value range. However, it is common for samples belonging to different groups to occupy the same regions of the embedding space. For example, the discretization step in Fig. 3.c assigns the samples in the top part of the embedding into three separate region annotations. However, ~~spatially,~~ these regions exhibit near-perfect spatial overlap, and upon inspecting their associated rules, we find that they ~~could~~ can be consolidated into a single region annotation. Merging these kinds of regions reduces the number of rules and regions, simplifying the resulting visual explanations and reducing the cognitive load on the end user.

~~In order to~~ To identify and consolidate overlapping and semantically related region annotations, we ~~develop~~ introduce the following post-hoc merging procedure. For a set of region annotations to be merged, they must satisfy two criteria: high spatial overlap and semantic compatibility between their rules.

Determining the semantic compatibility of rules is straightforward. For rules derived from ordinal or numeric features, only ~~consecutive~~ adjacent bins or categories may be merged. For rules originating from nominal categorical features, any combination of categories can be merged.

[1]These regions likely correspond to spurious artifacts of the dimensionality reduction process rather than any true underlying structure

Determining the degree of spatial overlap between two regions is similarly straightforward. We define the *maximum overlap* between two regions r_i and r_j as

$$\text{max_overlap}(r_i, r_j) = \max\left\{\frac{|r_i \cap r_j|}{|r_i|}, \frac{|r_j \cap r_i|}{|r_j|}\right\},$$

where $|r|$ denotes the number of samples within that region. A pair of regions ~~can be merged if~~ is merged when their maximum overlap exceeds a predetermined threshold. This formulation ensures that small regions, which are completely encompassed by a larger region, can be absorbed into the larger region.

In our running example, the three region annotations at the top of Fig. 3.c exhibit near-perfect overlap, and their associated rules are semantically compatible. Therefore, the three regions can be merged into a single region annotation, shown in Fig. 3.d.

Following the merging step, features with little or no spatial localization ~~will end up with~~ typically result in a single region annotation ~~,~~ covering the entire embedding and encompassing all feature values. This is because, for these features, their values are spread evenly across the embedding space, and the region construction step produces overlapping regions that extend across the entire embedding space, which the merging procedure promptly joins back up. Since ~~these~~ such singleton region annotations ~~provide no~~ do not convey meaningful information about the ~~structure of the embeddingspace~~embedding's structure, we discard them from subsequent analysis.

B. Contrastive Explanations

Contrastive explanations show how the values of a particular feature ~~are distributed~~ vary across the embedding space. While conceptually similar to coloring points ~~based on their feature values~~by feature value, our approach automatically filters out spatially ~~irrelevant~~ uninformative features and ranks the remaining ones ~~according to their informativeness in explaining the observed patterns~~by their explanatory power. This allows users to focus on the most relevant features without ~~having to manually sift through potentially large numbers of all available features~~manually sifting through a potentially large set of candidates.

1) Explanation Construction: Constructing contrastive visual explanations is straightforward: we ~~simply~~ include all region annotations e_i associated with a ~~particular~~ given feature f. However, even after filtering singleton region annotations, we may still be left with a large number of candidate explanations. To further reduce the number of candidates, we ~~develop a procedure to automatically identify and merge features with a high degree of~~ introduce an automatic procedure that identifies and merges features exhibiting strong spatial correlation.

The *contrastive merge* combines features ~~with near-perfectly spatially overlapping region annotations~~ whose region annotations exhibit near-perfect spatial overlap. Unlike the post-hoc merging procedure described in Section III-A3, which merges only region annotations derived from a single feature, the contrastive merge combines region annotations derived from multiple features. ~~In order to combine~~ To merge

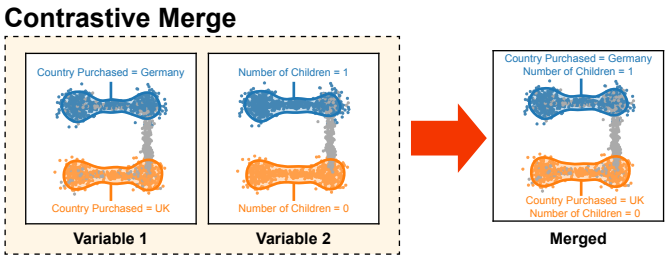


Fig. 4. The contrastive merge. If two features contain region annotations that have a nearly perfect overlap bipartite matching, we can merge the two features into one. This not only reduces the number of panels a user needs to inspect but also reveals relationships between features.

region annotations associated with two features f_1 and f_2, ~~we require that each associated~~ each region annotation $e_i \in f_1$ must have a nearly perfectly ~~spatially~~ overlapping counterpart $e_j \in f_2$. If such a bijection is found, we merge the matching pairs of region annotations, effectively representing two features using a single collection of region annotations. For example, the left panel in Fig. 4 depicts two separate features whose region annotations exhibit near-perfect overlap and can be merged into a single explanation shown on the right. This procedure not only reduces the number of candidate visual explanations but ~~describes more features using~~ also represents more features with fewer panels and reveals relationships ~~between~~ among high-dimensional features.

2) Explanation Selection: To identify ~~contrastive explanations most likely to be informative to end users~~the most informative contrastive explanations, we rank and select visual explanations ~~according to~~ using three criteria: ~~the~~ total region overlap, mean purity, and ~~the human attention~~ a human-attention score.

Visual explanations describing features whose values are localized to particular regions of the embedding space are likely to be more informative than features whose values are spread more evenly across the embedding space. Region annotations associated with highly localized features exhibit ~~lower overlap between their regions than their~~ less overlap than those of more evenly distributed ~~counterparts.~~ features.

We compute the overlap between two region annotations as the Jaccard similarity of their regions and define the *total region overlap* of a panel P by taking the mean overlap between all pairs of region annotations (e_i, e_j).

The *purity* of each region annotation indicates what proportion of the samples encompassed by the region satisfy its associated rule:

$$\text{purity}(r) = \frac{1}{|r|}\sum_{y_i \in r} \varphi_i(y_i).$$

High-purity regions ~~signify a stronger spatial pattern than low-purity regions and should be~~ indicate stronger spatial structure and are therefore preferred. For each panel, we compute the *mean purity* of the contained explanatory variables.

A well-known finding from psychology states that human working memory has limited capacity and can pay attention to only a handful of items at a time. Several bounds have been proposed, including seven [42], four [43], and two [44]. Panels

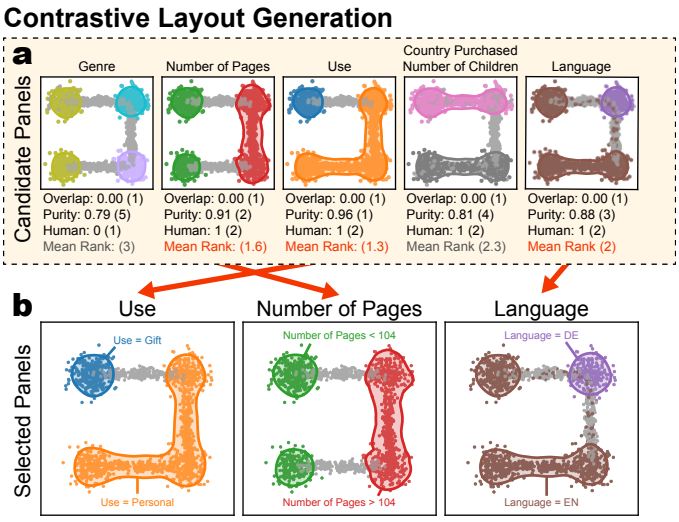


Fig. 5. Contrastive layout generation. (a) Candidate panels are generated by including all region annotations associated with a particular feature. We score and rank each candidate panel according to three metrics and obtain a final, weighted mean rank. (b) The top k panels, indicated in red, are included in the resulting layout.

containing ~~large numbers of~~ many region annotations are more likely to overwhelm ~~the user~~users; therefore, panels with fewer ~~region annotations should be~~ annotations are preferred. We define the ~~*human attention*~~ *human-attention score* ~~$|3 - |P||$~~ as $|3 - |P||$, where $|P|$ denotes the number of ~~contained~~ region annotations in panel P~~and assigns lower scores~~, assigning the lowest score to panels containing three ~~region~~ annotations. In our case, three ~~serves as our~~ represents an intermediate estimate of ~~how many~~ the number of items a user can comfortably ~~reason about at the same time~~consider simultaneously.

We compute individual panel rankings for each of ~~these~~ the three metrics and ~~compute their~~ aggregate them using a weighted mean to obtain ~~a~~ the final ranking. We then select the top k panels with the highest overall ranking as our final contrastive explanations. This process is illustrated in Fig. 5.

C. Descriptive Explanations

Descriptive explanations ~~describe~~ characterize regions of the embedding space by ~~determining~~ identifying sets of features and ~~their~~ feature values that are characteristic of the samples ~~contained within these~~ within those regions. Unlike contrastive explanations, which primarily show how the values of a single feature ~~are distributed across different~~ vary across regions, descriptive explanations ~~describe regions in terms of combinations of multiple features~~ capture how multiple features jointly characterize specific regions.

1) Explanation Construction: To identify candidate regions and their characteristic feature values, our ~~procedure finds~~ method detects sets of region annotations ~~exhibiting~~ with a high degree of overlap and merges them into ~~region~~ annotation groups. Each resulting group then describes a particular region using the combination of features and feature values of each included region annotation. As in contrastive explanations, we use the region annotations produced by the region construction procedure outlined in Section III-A as our starting point.

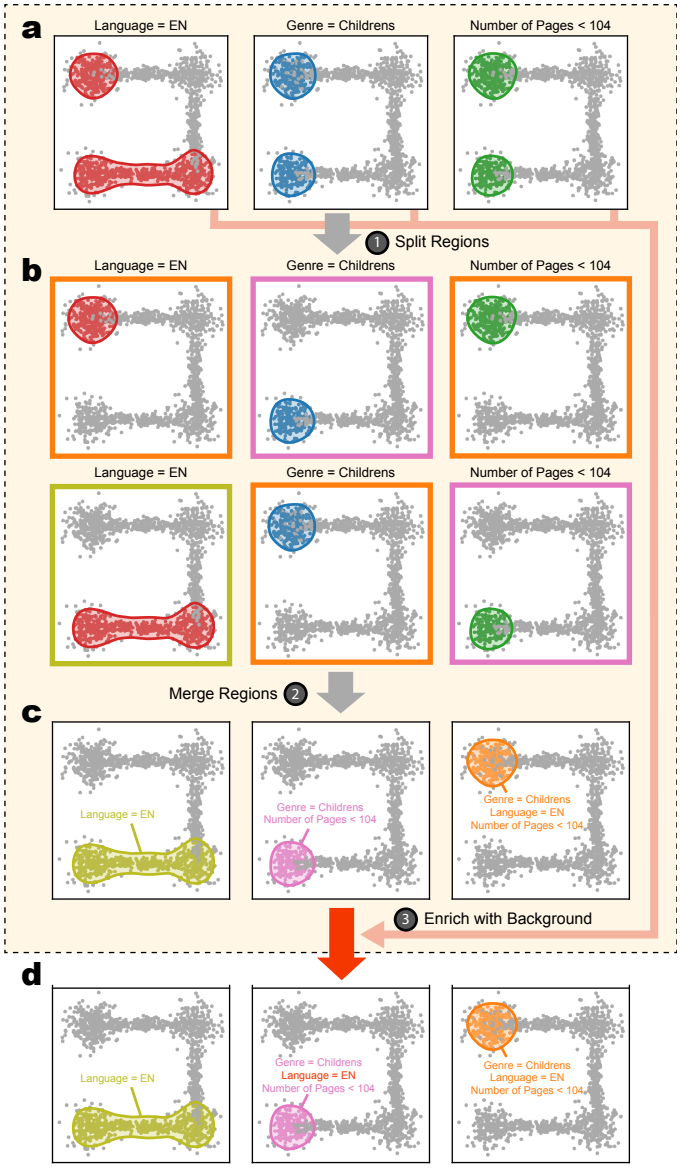


Fig. 6. The descriptive merge, demonstrated on three region annotations (a). First, we split region annotations into disjoint regions (b). Then, we identify spatially overlapping region annotations and merge them into region annotation groups (c). Finally, each region annotation group is enriched with background annotations (d). In this instance, the middle panel depicting the pink region receives an additional descriptor, highlighted in red.

To ~~find groups of~~ group overlapping region annotations, we ~~devise~~ introduce the *descriptive merge*, ~~comprised of the three steps outlined~~ which consists of three main steps illustrated in Fig. 6. First, we split region annotations containing multiple disjoint regions into separate region annotations. For instance, the three region annotations shown in Fig. 6.a each contain two disjoint regions, ~~each of which is split into two region~~ which are split into separate annotations in Fig. 6.b.

Second, we merge sets of overlapping region annotations into region annotations groups. Two region annotations can be merged if their regions r_i and r_j exhibit a sufficiently high

degree of *minimum overlap*:

$$\text{min_overlap}(r_i, r_j) = \min\left\{\frac{|r_i \cap r_j|}{|r_i|}, \frac{|r_j \cap r_i|}{|r_j|}\right\}.$$

Third, while the previous step merges region annotations exhibiting near-perfect overlap, these merged region annotations may not fully describe the characteristics of the samples within the region. For instance, the pink region in the middle panel of Fig. 6.c represents short children's books, obtained by merging the near-perfectly overlapping regions in Fig. 6.b highlighted in pink. However, these books are also fully contained within the yellow region corresponding to the English language in the left panel of Fig. 6.c. ~~Therefore, it would be~~ Hence, it is more informative to describe ~~these books~~ them as short, English children's books, as shown in Fig. 6.d, rather than ~~just~~ simply as short children's books. To incorporate these additional characteristics, the third step enriches the generated region annotations with rules from larger, sufficiently overlapping regions. Importantly, we do not merge the regions themselves; ~~we only add rules from the largerregions to the smaller~~ instead, we augment smaller regions with rules from larger, overlapping ones. We refer to this process as *background enrichment*.

Step 2 ~~is reminiscent of~~ resembles the post-hoc merging procedure described in Section III-A3, ~~where we required~~ in which the *maximum overlap* had to exceed a ~~pre-specified~~ predefined threshold. There, the maximum overlap ensured that smaller region annotations could be absorbed into larger, encompassing regions. Here, however, this approach would lead to misleading annotations. Consider merging two region annotations e_i and e_j associated with rules φ_i, φ_j and regions r_i, r_j respectively. If r_j is a small region entirely contained within the larger region r_i, the points in r_j will satisfy both rules φ_i, φ_j, while the remaining points in r_i that are not in r_j will not satisfy φ_j but only φ_i. However, the merged rule φ_{new} would ~~indicate that the~~ incorrectly indicate that all points within the merged region satisfy both ~~rules~~ φ_i and φ_j, ~~resulting in~~ leading to a misleading annotation. To avoid this, we ~~here~~ instead require a high degree of *minimum overlap* between two regions, ensuring that both ~~regions cover roughly~~ cover approximately the same data points and that the resulting annotation remains valid ~~for all the encompassed data points~~across all encompassed samples.

2) Explanation Selection: Descriptive explanations are ~~obtained in an iterative fashion~~generated iteratively. Given an initial pool of available region annotations, we identify all their potential combinations such that the overlap between them does not exceed some user-specified threshold[2]. Each candidate panel is scored across five criteria. The resulting scores are ranked and aggregated using a mean-rank strategy, similar to that in Section III-B2, and the highest-ranking panel is ~~included in~~ added to the final layout. The region annotations from the selected panel are then removed from the candidate pool, and the procedure is repeated until we generate the requested number of panels k or until the pool of available region annotations is exhausted. This procedure is illustrated in Fig. 7.

~~In order to~~ To efficiently identify combinations of non-overlapping region annotations, we formulate ~~this~~ the task as an ~~independent set~~ independent-set problem. We represent each region annotation as a node in a graph and place edges between them if their overlap exceeds a predetermined threshold. Then, the independent sets of this graph correspond to sets of non-overlapping region annotations, which are then considered candidate panels.

Descriptive explanations are ranked and selected based on five criteria. The first three include the *sample coverage*, i.e., what proportion of all data points are covered by the region annotations in the panel, the *mean purity*, and the *human attention score* from Section III-B2. These metrics favor panels that offer broad coverage of the embedding space, contain reliable regions, and remain visually manageable in terms of the number of annotations. The fourth and fifth criteria ensure a degree of contrastivity, which, according to Miller [38], should lead to more intuitive and satisfying explanations. Although descriptive explanations ~~can characterize regions in terms of different, disjoint~~ may characterize regions using distinct sets of features, we prioritize panels ~~in which features appear~~ where features recur across multiple region annotations. This ~~allows~~ enables users to compare feature values across ~~different~~ regions within a single panel~~rather than having~~, without needing to reference other visual explanations. To prioritize feature repetition, we score each panel according to its *mean variable occurrence* (MVO), defined as the average number of times each feature appears across all region annotations within a particular panel. Panels with higher MVO scores are preferred, as they facilitate within-panel comparisons. We count the number of features that appear in all region annotations within a given panel using a dedicated *variable in all* (VIA) score, favoring panels with higher VIA scores. Together, these five metrics ensure ~~that we identify~~ the selection of high-quality region annotations that comprehensively cover the embedding space ~~and that their contents align with human preferences and cognitive abilities~~while aligning with human perceptual and cognitive preferences.

IV. Usage Scenarios

To demonstrate the versatility and effectiveness of VERA, we present ~~the~~ generated explanations across several diverse datasets.

A. IBM Employee Attrition Dataset

We begin with a detailed analysis of the IBM Employee Attrition dataset [45]~~demonstrating~~, illustrating how VERA explanations ~~can help understand the visual patterns in a given~~ aid in interpreting visual patterns within a two-dimensional embedding. The ~~data set describes~~ dataset contains records of 1,470 fictional employees, each characterized by 33 features. We obtain a two-dimensional embedding ~~using~~ with the t-SNE algorithm ~~using~~ implemented in the openTSNE library [46].

[2]Unlike in the merging procedure, in which overlapping region annotations were merged together, a high degree of overlap here would result in overlapping annotations in the final visualization, which would render the visualization unreadable.

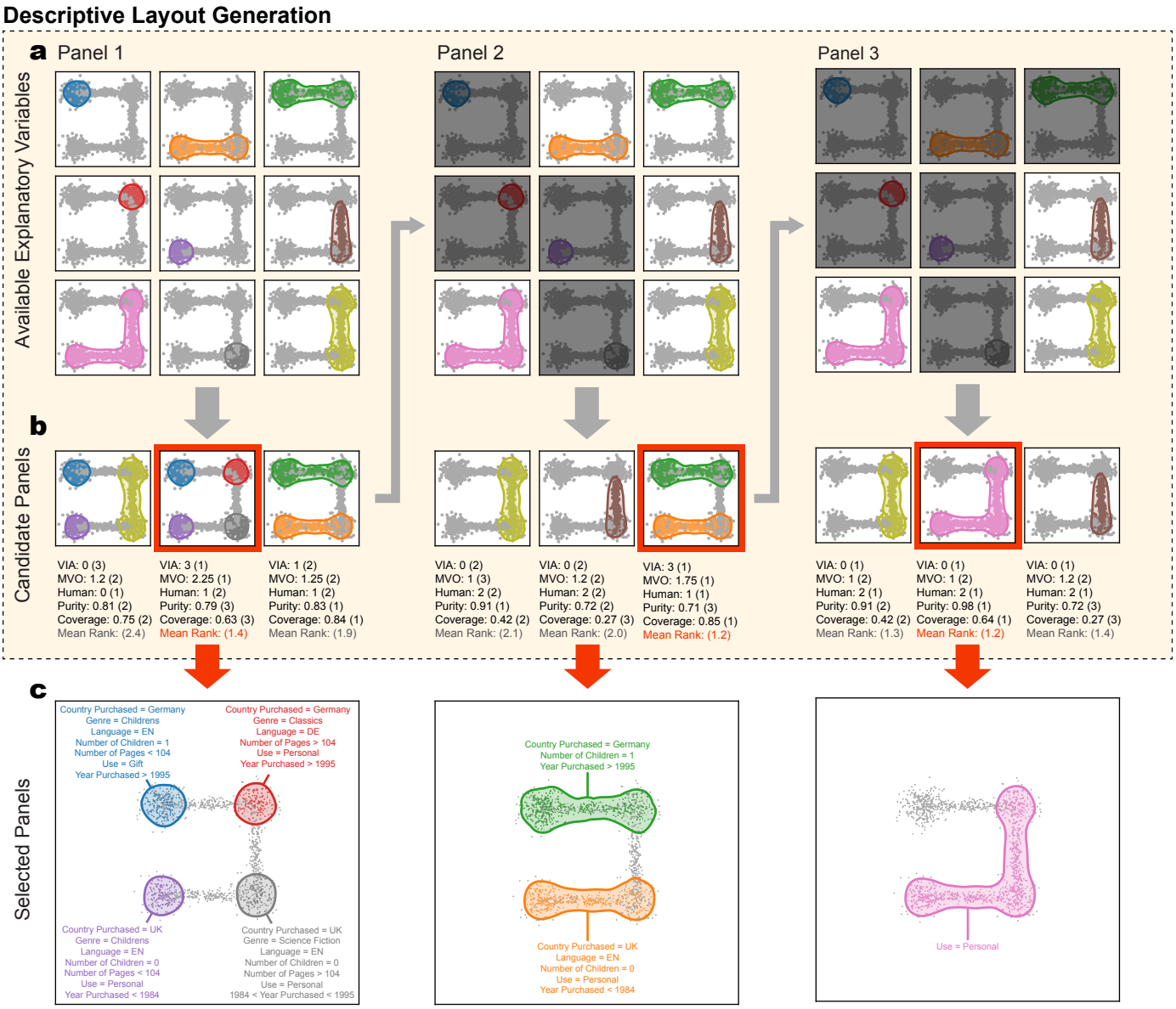


Fig. 7. The descriptive layout generation follows an iterative approach: given an initial pool of region annotations in (a), we generate candidate panels by identifying their non-overlapping combinations. We show three such candidate panels in (b). Then, we score each panel along five different metrics, which are subsequently ranked and averaged. The panel with the highest mean rank, indicated in red, is added to the resulting layout, shown in (c). Finally, we remove the region annotations in the selected panel from the candidate pool. We then repeat the procedure until we obtain a specified number of panels or until the pool runs out of region annotations.

Fig. 8 shows the ~~top four~~ four highest-ranked contrastive and descriptive explanations selected by VERA. The first two contrastive explanations in Figs. 8.a and b reveal that the three clusters are primarily distinguished by employees' departments and education fields. Fig. 8.c shows that the three clusters can be further subdivided into specific job roles, while Fig. 8.d shows that the number of total working years also corresponds to the samples' spatial positions. These four panels alone already offer a meaningful interpretation of the three main clusters uncovered by the t-SNE algorithm.

The descriptive explanations depicted in Figs. 8.e-h offer descriptions of specific regions of the embedding space. For instance, the blue cluster in Fig. 8.e lists the characteristics of the human resources department and primarily includes junior, married employees with lower salaries. In contrast, the red cluster contains more senior, married individuals with higher salaries. Cross-referencing ~~this~~ Fig. 8.c ~~, we can see that most of the~~ reveals that most employees in the red cluster hold managerial or research director roles. We can similarly inspect the other regions to gain further insight into the structure captured by the embedding.

Notice that many of the identified regions in Fig. 8 do not correspond to discrete clusters but rather represent regions within or across clusters. For instance, the gray region in Fig. 8~~.~~g corresponds to single employees with less than 15 years of experience and low job levels~~.~~ ~~—~~ likely younger, junior staff. ~~On the other hand~~Conversely, the yellow region primarily includes divorced employees~~. We are provided with~~

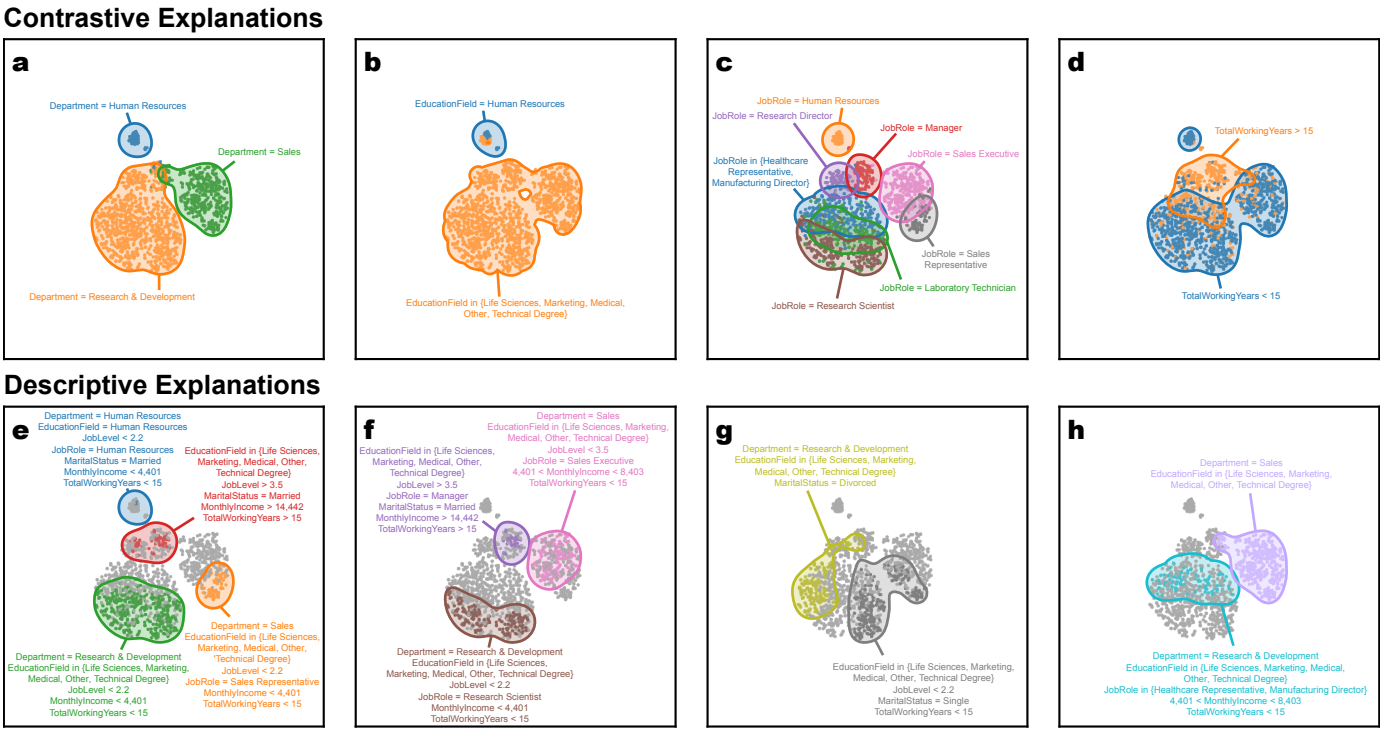


Fig. 8. Explanations of the IBM Employee Attrition data set. Panels (a-d) show four contrastive explanations corresponding to the spatially most correlated features with this particular embedding. The descriptive explanations shown in panels (e-h) characterize different regions of the embedding space.

~~no information about their job level or~~ , for whom job level and total working years ~~. Therefore, we do not assume any particular values about these employees. This shows VERA'~~are unspecified. This demonstrates VERA's ability to ~~not only characterize~~ characterize not only entire clusters but ~~potentially interesting parts of clusters~~also informative subregions within them.

B. Abalone Dataset

The Abalone dataset [47] is a well-known benchmark in regression tasks, containing 4,177 samples of abalones – marine mollusks with distinctive ear-shaped shells. Each sample is characterized by eight features, including physical measurements of the body and shell ~~, as well as~~ and biological attributes such as age and sex. We generate a two-dimensional embedding with the UMAP algorithm using the umap-learn library [7]. The ~~obtained~~ resulting embedding reveals no distinct clusters but forms an elongated ~~shape stretching~~ structure extending from left to right.

Fig. 9 shows four selected contrastive explanations generated by VERA. The explanations reveal that the elongated shape reflects the size and age of the abalones: younger abalones are smaller and lighter, while older abalones are larger and heavier. Fig. 9.b also suggests that the sex of the animals does not affect their size. Figs. 9.c-d highlight the benefits of the contrastive merge procedure. Fig. 9.c combines shucked weight, viscera weight, and whole weight into a single explanation, while Fig. 9.d combines shell diameter and length. This example illustrates how VERA can reduce the number of feature explanations analysts need to examine – in this case, from five to two – while simultaneously revealing the strong correlation between these three features.

C. Iris and Penguin Datasets

The Iris [48] and Penguin [49] datasets are widely used benchmarks in data visualization and machine learning due to their clear clustering structure and overall simplicity. Fig. 10 shows one descriptive explanation for each data set, ~~each of which provides a complete~~ providing a concise overview of the main patterns and relationships. In the Iris dataset (Fig. 10.a), VERA identifies regions corresponding to the three species of iris flowers, each characterized by distinct combinations of sepal and petal measurements. For the Penguin dataset (Fig. 10.b), VERA identifies clusters representing the three penguin species, separated by features such as flipper length and body mass. For simple ~~data sets~~ datasets such as these containing relatively few features and having relatively well-defined clusters, a single descriptive explanation ~~can be sufficient to fully convey the~~ is often sufficient to convey the entire embedding structure.

V. User Evaluation

To evaluate the effectiveness of our approach, we conducted a ~~comparative~~ user study comparing ~~the~~ participants' speed and accuracy ~~with which users could reason~~ in reasoning about low-dimensional embeddings using ~~both an interactive data exploration tool and our newly proposed~~ an interactive exploration tool versus our static VERA explanations.

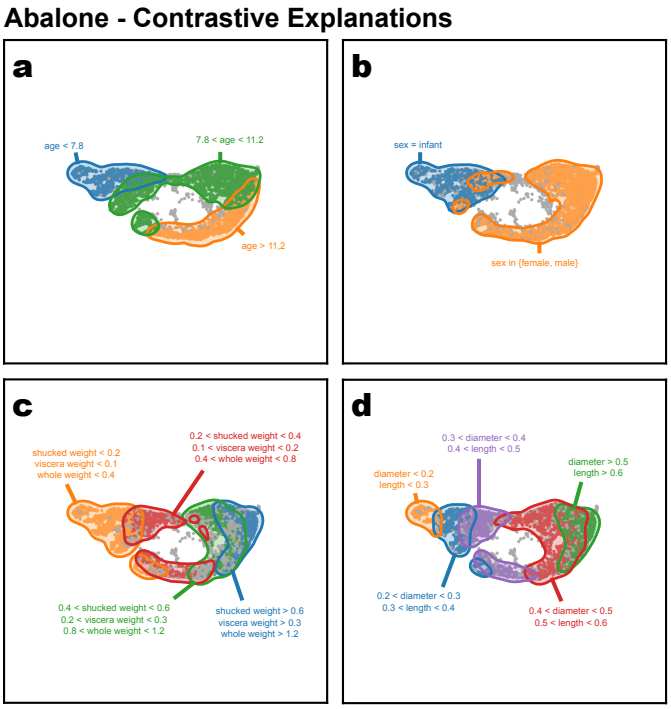


Fig. 9. Contrastive explanations for a UMAP embedding of the Abalone dataset. Panels (a-d) highlight selected features, showing the spatial distribution of age, sex, and physical measurements, as well as correlations between features related to animal weight and size.

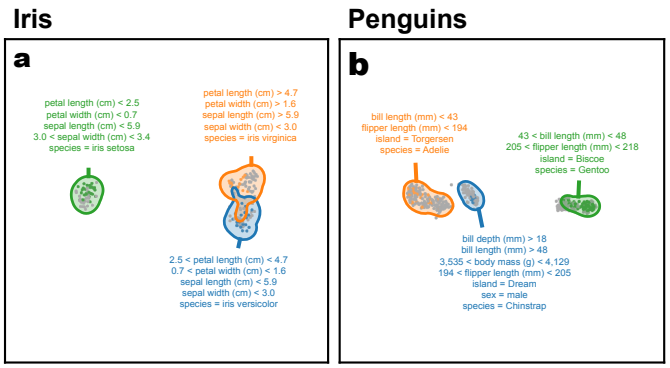


Fig. 10. Descriptive explanations of the Iris (a) and Penguin (b) ~~data sets~~datasets. Each explanation provides a comprehensive, high-level overview of the clusters revealed in the embedding.

A. Study Design

We ~~select~~ selected the Orange data mining toolkit [50] as our interactive tool of choice because of its ease of use and focus on visual data exploration. ~~While~~ Although not explicitly designed for interpreting embeddings, Orange includes standard visualization tools for exploring two-dimensional embeddings. Many related approaches based on feature distribution plots [19], [39] and discriminative models [22], [36] can be easily reproduced in Orange using only a handful of widgets.

We ~~prepare two sets~~ prepared two groups of three real-world ~~data sets each, with varying~~ datasets each, selected to vary in complexity. The first data set group D_1 comprises the Iris [48], Employee Attrition [45], and Raisin [51] data sets. The second data set group D_2 includes the Penguins [49], Titanic [52], and Wine Quality [53] data sets. For each data set, participants ~~were asked to answer~~ answered a series of typical ~~questions arising during the exploration of~~ exploratory questions about two-dimensional embeddings, ~~e.g.,~~ such as: "What makes this cluster different from the remaining clusters?", "What characterizes the samples in this cluster?", and "Do you notice any trend along this region?".

We recruited 57 bachelor's students enrolled in the Introduction to Machine Learning course at the Faculty of Computer and Information Science at the University of Ljubljana. Participation was purely voluntary, and no personally identifiable information was collected. Prior to the study, ~~the participants had some familiarity~~ participants were already familiar with Orange but ~~were not familiar~~ not with VERA explanations. ~~Before conducting the study, we provided a short~~ We provided a brief tutorial on both ~~Orange and VERA explanations. We randomly assigned participants~~ tools before beginning the experiment. Participants were randomly assigned to one of two ~~user~~ experimental groups. The first group G_1 first used Orange to answer questions related to datasets D_1, then used VERA explanations for questions relating to datasets in D_2. Conversely, ~~the second~~ group G_2 first used VERA ~~to answer questions relating to~~ on D_1 ~~, then used Orange to answer questions relating to the data sets in~~ and then Orange on D_2. On average, students answered 29.2 of the 32 questions (SD $=$ 4.3, SE $=$ 0.56), amounting to a total of 1,666 responses.

To assess ~~the participant's~~ response time and accuracy, we ~~timed how long it took each participant~~ measured how long each participant took to answer each question ~~related to a particular data set group and determined whether they answered the questions correctly. Upon completing both data set groups, we asked participants to answer~~ and whether the response was correct. After completing both dataset groups, participants filled out a final questionnaire ~~in which they could provide feedback related to the two~~ to provide qualitative feedback on both tools.

B. Results

The results ~~of our study~~ show that participants using VERA explanations performed as ~~well as participants using Orange, but taking~~ accurately as those using Orange but required significantly less time. The difference in user performance was not statistically significant (~~t-test~~ t-test $p = 0.084$), with 91% (SE $= 0.9\%$) of participants answering correctly when using Orange and 93% (SE $= 1.0\%$) when using VERA. ~~On the other hand~~However, participants in both user groups answered ~~questions~~ significantly faster when using VERA explanations ($p < 0.005$). On average, participants using Orange took 29 minutes (SD $= 11.5$, SE $= 2.1$) to complete questions related to D_1 and 23 minutes (SD $= 6.0$, SE $= 1.1$) to complete questions related to D_2. Conversely, participants using VERA explanations required 21 minutes (SD $= 6.3$, SE $= 1.2$) and 15 minutes (SD $= 5.4$, SE $= 1.1$) for each data set group, representing a consistent increase in speed of approximately 33%. These results confirm that participants in both groups were able to answer the questions with roughly the same

degree of accuracy but at a much faster rate when using VERA explanations.

Interestingly, when asked to list characteristic or distinguishing features for particular regions, participants often mentioned more features when using VERA explanations ($p = 0.04$). When generating explanations using Orange, participants mentioned an average of 2.0 (SD $= 1.29$, SE $= 0.07$) features in their answers, while this increased to 2.2 (SD $= 1.33$, SE $= 0.07$) when using VERA explanations. This ~~discrepancy could be due to~~ difference may stem from several factors. ~~For example, when exploring two-dimensional embeddings with~~ When using an interactive tool, users ~~typically have to manually sift through the many different features to explore their correspondence~~ must manually inspect multiple features to assess their relationship to the embedding space. Participants often ~~stop~~ stopped their exploration after identifying one or two informative features, potentially ~~missing other equally informative features~~overlooking additional relevant ones. VERA explanations, on the other hand, provide all relevant explanations at once. ~~While these explanations are limited in the~~ Although these explanations display only a limited number of features~~they display, the explanation ranking mechanisms in VERAensure that the user sees the most salient features~~ , VERA's ranking mechanism ensures that the most salient ones appear first. Thus, the user simply *sees* all the features and cannot miss them by accident.

In the final feedback questionnaire, we asked participants to document their experience with both tools. Consistent with our quantitative results, most participants (91%, SE $= 4\%$) reported that they were able to answer questions more quickly using the VERA explanations. The majority of participants (77%, SE $= 6\%$) also confirmed that the static explanations ~~provided were just~~ were as informative as those produced with Orange. Surprisingly, when asked which tools they trusted more, 50% (SE $= 8\%$) of participants said they trusted both tools equally, 20% (SE $= 6\%$) said they trusted VERA explanations more, and 30% said they trusted Orange more (SE $= 7\%$). We ~~expected~~ anticipated that users would ~~tend to~~ trust full-featured, interactive toolkits like Orange more than static visual explanations~~because these tools provide users with multiple methods to verify their findings and allow them~~ , as interactive tools enable multiple validation mechanisms and allow users to drill down into specific ~~parts of the visualization~~ visualization components. VERA explanations, on the other hand, are static and offer no such options, forcing the user to simply trust the resulting visualizations.

Finally, ~~we asked participants which~~ participants rated the usefulness of both types of VERA explanations ~~they found more informative and asked them to rate each type of explanation for its usefulness on a typical~~ on a five-point Likert scale and indicated which type they found more informative. Consistent with Miller's [38] findings, most participants (75%, SE $= 6.6\%$) selected contrastive explanations as the most informative, while 15% (SE $= 5.4\%$) found both types of explanations equally informative, and 10% (SE $= 4.7\%$) found descriptive explanations more informative. Contrastive explanations received a higher overall rating of 4.3 (SD $= 0.89$, SE $= 0.13$), while descriptive explanations received a lower overall rating of 3.2 (SD $= 1.06$, SE $= 0.16$). ~~We note that this~~ This result does not ~~imply~~ suggest that descriptive explanations ~~are not useful but merely~~ lack utility. Rather, it indicates that users generally ~~find contrastive explanations more informative than descriptive explanations~~perceive contrastive explanations as more informative.

VI. Scalability and Limitations

~~When considering scalability , we consider~~ We assess scalability from two perspectives, algorithmic complexity and visual limitations.

Algorithmic complexity can be ~~analysed in terms of~~ analyzed with respect to the number of samples N and ~~the number of~~ features F. In VERA, the samples are used only during the region construction step described in Section III-A, where ~~we obtain the KDE~~ a kernel density estimate is computed for each feature. ~~A~~ The KDE can be efficiently computed in linear time using an FFT-accelerated interpolation-based approach, ~~resulting in~~ yielding a linear time complexity $\mathcal{O}(N)$. Since the remaining steps of VERA operate solely on the extracted region annotations, the overall time complexity with respect to N is linear, reflected in Fig. 11.a. ~~We also note that accurate estimates of the KDE can be obtained by subsampling the dataset~~ Accurate KDE estimates can also be obtained through dataset subsampling, further improving scalability.

The number of features F has a greater impact on runtime. After extracting ~~the~~ initial region annotations, VERA ~~applies~~ performs a post-hoc merging ~~procedure to annotations from~~ step for each feature, ~~which takes~~ requiring linear time $\mathcal{O}(F)$. Both contrastive and descriptive explanations ~~then~~ construct an overlap graph of ~~the resulting region annotations , which involves calculating the pairwise overlap between all pairs of region annotations, requiring $\mathcal{O}(F^2)$ time~~ region annotations by computing pairwise overlaps, an operation with quadratic time complexity $\mathcal{O}(F^2)$. The overlap graph is then used for both contrastive and descriptive merge steps. In contrastive explanations, the remaining step is a linear time panel ranking, resulting in an overall time complexity of $\mathcal{O}(F^2)$, as can be seen in Fig. 11.b. In descriptive explanations, we use the merged region annotations to construct a second overlap graph and use its independent sets as candidate panels. Finding independent sets is NP-hard, requiring $\mathcal{O}(2^F)$ time. ~~However, the actual runtime of our procedure is often~~ In practice, runtime is typically much lower and ~~can vary significantly~~ varies across datasets, as ~~illustrated~~ shown in Fig. 11.b. This ~~is due to~~ reduction stems from the descriptive merge step, which often ~~greatly reduces the total~~ substantially decreases the number of candidate region annotations, ~~resulting in a much smaller graph and , consequently, lower independent set runtime~~yielding smaller graphs and faster independent set computations. For datasets with ~~a large number of features~~F, ~~we have found that greedy graph coloring results in~~ many features, a greedy graph-coloring heuristic yields good approximate solutions ~~and requires only~~ in linear time. In this case, ~~descriptive explanation generation is dominated by the graph construction~~ graph construction ($\mathcal{O}(F^2)$) dominates the descriptive explanation generation time. Although both

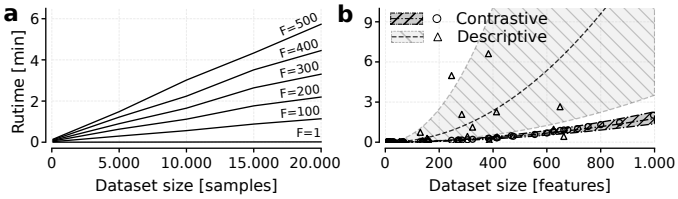


Fig. 11. ~~We benchmark the single-core~~ Single-core performance of VERA on a consumer-grade M1 MacBook Pro on different-sized synthetic datasets. (a) VERA scales linearly in the number of samples N, as shown by the six near-straight lines, corresponding to datasets with a different number of features F. (b) Runtime with respect to the number of region annotations (after post-hoc merging and filtering) F for contrastive and descriptive explanation generation.

~~contrastive and descriptive explanations exhibit superlinear time~~ explanation types exhibit superlinear complexity, VERA remains practical for ~~many~~ typical real-world datasets, ~~with execution times ranging from~~ completing in seconds to a few minutes on standard hardware.

Visual limitations primarily ~~affects~~ affect descriptive explanations, ~~particularly when the number of features~~ especially when the feature count is large. In these cases, VERA may produce long lists of region descriptors – particularly when background enrichment is used – which pose two key challenges. First, long descriptor lists impose a cognitive burden on users, who must manually inspect and interpret the output. Second, these lists consume screen space, leading to visual clutter: labels may overlap with each other or with data points, extend off-screen, or otherwise obstruct the view. Future work could ~~mitigate~~ address these issues by leveraging large language models (LLMs) to ~~generate short~~automatically generate concise, informative summaries of ~~long~~ lengthy region descriptors.

VII. Conclusion

With the growing popularity of two-dimensional embeddings ~~for~~ in data visualization and their ~~ubiquity during~~ widespread use in exploratory data analysis and ~~science communication,~~ scientific communication, there is a growing need for complementary tools to interpret, validate, and compare ~~such embeddingsmust also be developed. Here, we~~ these embeddings. We introduced *Visual Explanations via Region Annotation (VERA)*, ~~a novel automated embedding annotation tool designed to help users quickly~~ an automated method that enables users to rapidly interpret two-dimensional embeddings through static visual explanations. ~~Our approach leans~~ VERA draws on insights from the social sciences to design ~~highly informative visualizations that align closely~~ visual explanations that align with human cognitive processes~~and generates explanations~~, producing outputs that are both informative and intuitively satisfying. VERA addresses several ~~key limitations of current~~ limitations of existing interactive and static ~~approaches, both retaining the speed~~ methods, combining the efficiency of static ~~approaches while at the same time remaining as informative as its comprehensive interactive counterparts on many~~ visualization with the interpretive richness of interactive exploration in common exploratory data analysis tasks. Despite its effectiveness, ~~our work could be extended in several important ways and provides opportunities~~ VERA opens several promising directions for future research. Extending VERA to handle diverse data modalities, including image, video, text, and graph data, addressing scalability with large feature sets, and exploring integration with interactive analytics platforms all represent promising directions. Furthermore, recent advances in LLMs for automatic summarization could be leveraged to generate more concise and relevant annotations. Overall, VERA represents a ~~meaningful step towards a fastand effectiveapproach for explaining~~ significant step toward fast, effective, and cognitively aligned explanations of two-dimensional non-linear embeddings.

Code Availability

Our open-source, Python ~~implentation~~ implementation of VERA is ~~freely-available~~ freely available at https://github.com/pavlin-policar/vera and is licensed under the ~~BSD-3~~ BSD 3-clause license. The materials used to prepare this manuscript are available at https://github.com/pavlin-policar/vera-paper.

Acknowledgments

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